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Memory Reactivation Decoder

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Signature: Nitzan Censor



Project Topic & Motivation

Glossary

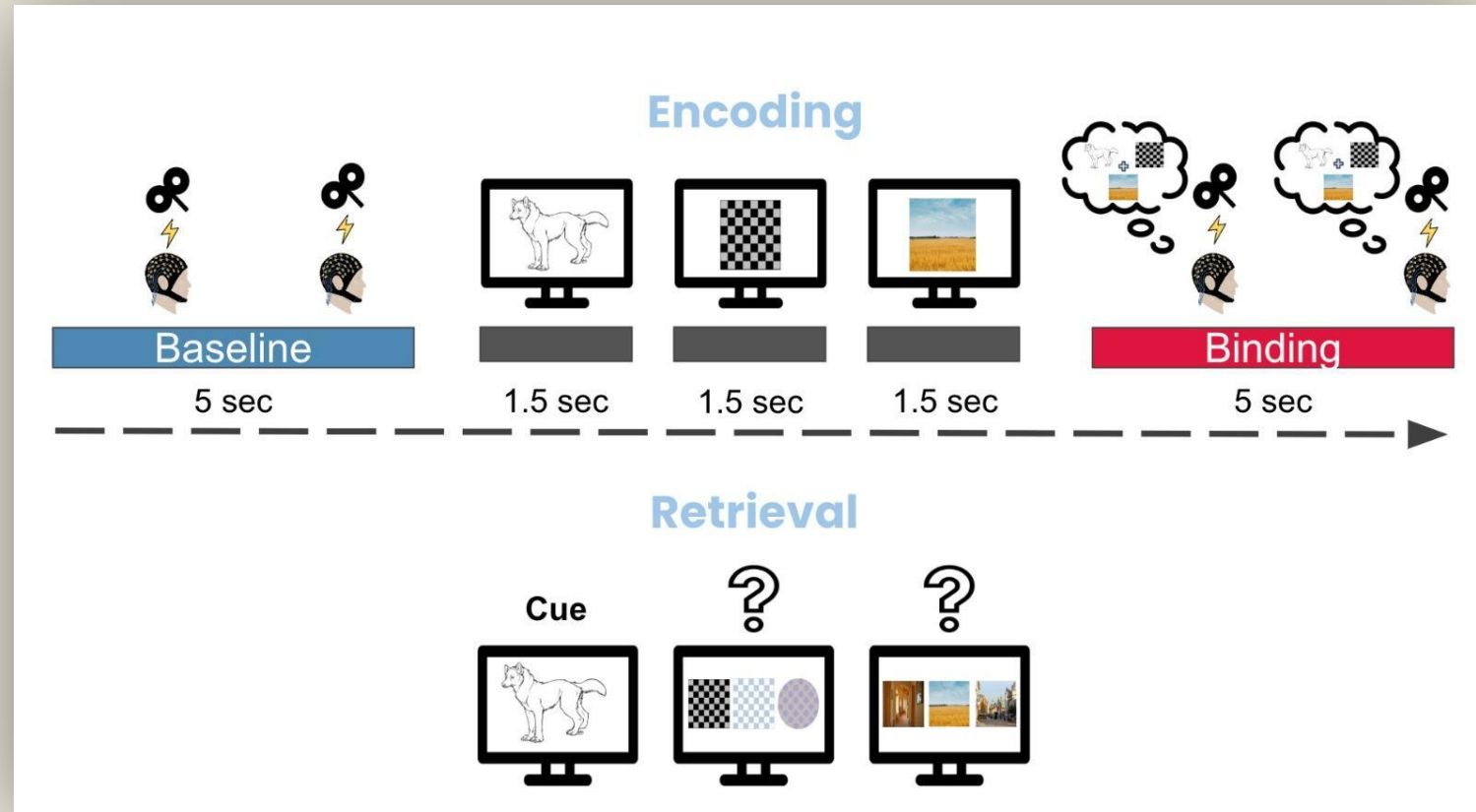
- **Memory Reactivation:**
The sequential replay of experience-specific neural patterns. Fundamental for episodic memory retrieval and long-term consolidation.
- **Retrieval:**
A process initiated by a cue that results in reactivation and episodic remembering.
- **Reconsolidation:** A memory process that occurs after a memory is retrieved, in which previously consolidated memories can be made labile again.

Project Goal

- Develop a **Memory Reactivation Decoder**: A machine-learning system to detect these events in real-time.
- Enable "**Action-Driven Decoding**" for future closed-loop interventions (e.g., targeted stimulation).
- Potential Use Cases:
 - Researching spontaneous reactivation in the brain.
 - Real-time interference with memory reactivation.

Experimental Data

Based on pre-recorded EEG data from a lab experiment where subjects learned associations between triads of visual stimuli.



Note: Data contains artifacts and noise as the experiment was not originally designed for our specific goal.



System Methodology

One-vs-All Decoders

Implementing a new classifier strategy starting from a functional localizer and identifying the correct category on retrieval.

(Transitioned from animate vs inanimate strategy)

Two-Stage Implementation

Offline Stage: Evaluate different models for all timepoints on offline data, selecting best performing models and peak reactivation time point.

Online Stage Adaptation: Streaming and processing EEG data, enabling to identify reactivation in real time lab trials.



Deliverables & Requirements

Offline Package

- Feature Extraction
- Model training, evaluation and selection
- Inference on offline data for evaluation of decoder performance

Requirements:

- **Decoder Performance $\geq 70\%$:** Indicates a 70% probability that the decoder will correctly distinguish a target neural signature from a non-target stimulus when comparing a random pair of epochs.
(Derived from AUC)

Online Package

- Real-time pipeline
- Action-Driven Logic for digital triggers.
- UI for online system

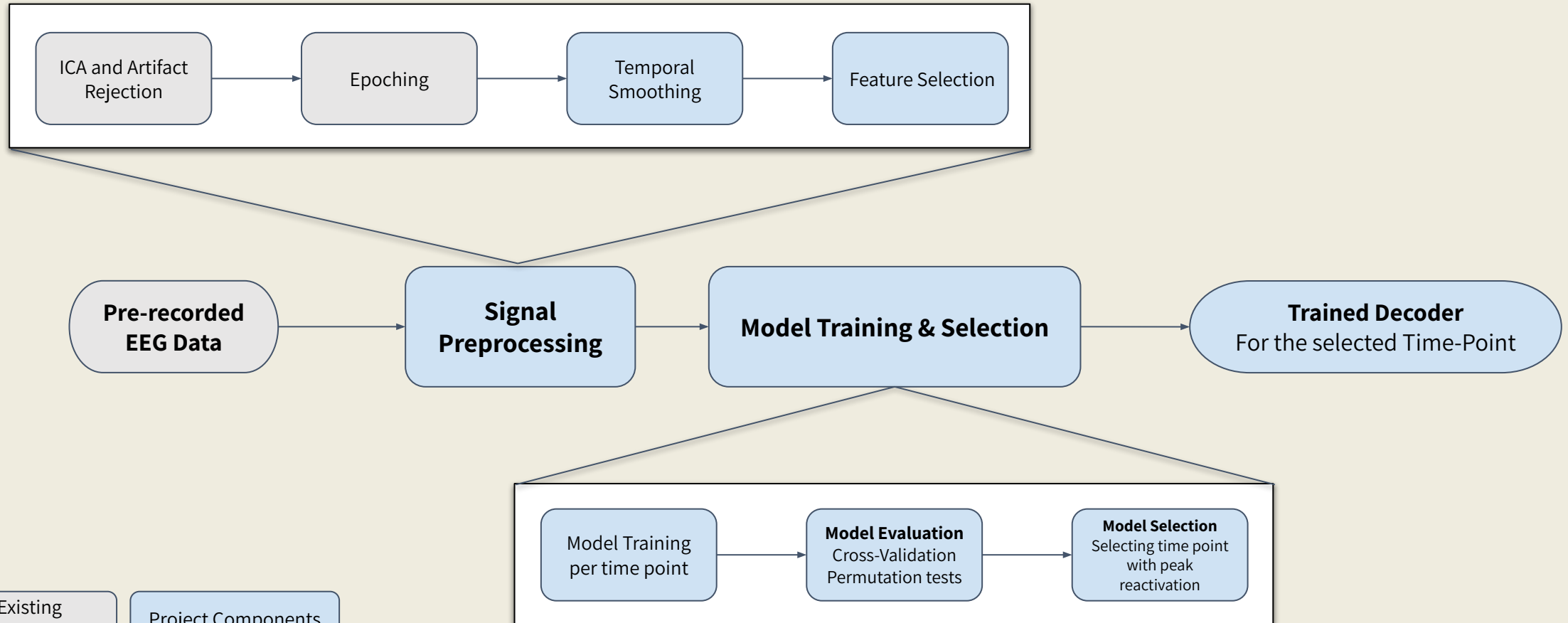
Requirements:

- **System Latency $\leq 100\text{ms}$:** The time between data acquisition and the final prediction of the object category must remain under 100ms.
- **Performance Drop $\leq 10\%$:** The live trained decoders evaluated using the same metric from the offline stage must exceed 60%.



Block Diagram – Offline Pipeline

Goal: Train Decoder for identifying reactivation in offline data



Pre-Existing
Components

Project Components

All components were implemented in semester A

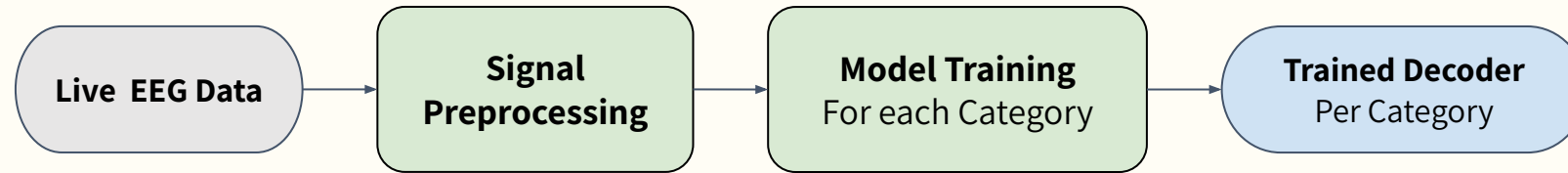


Block Diagram – Online Pipeline

Goal: Training decoders during lab trials and using them to identify reactivation with live subjects

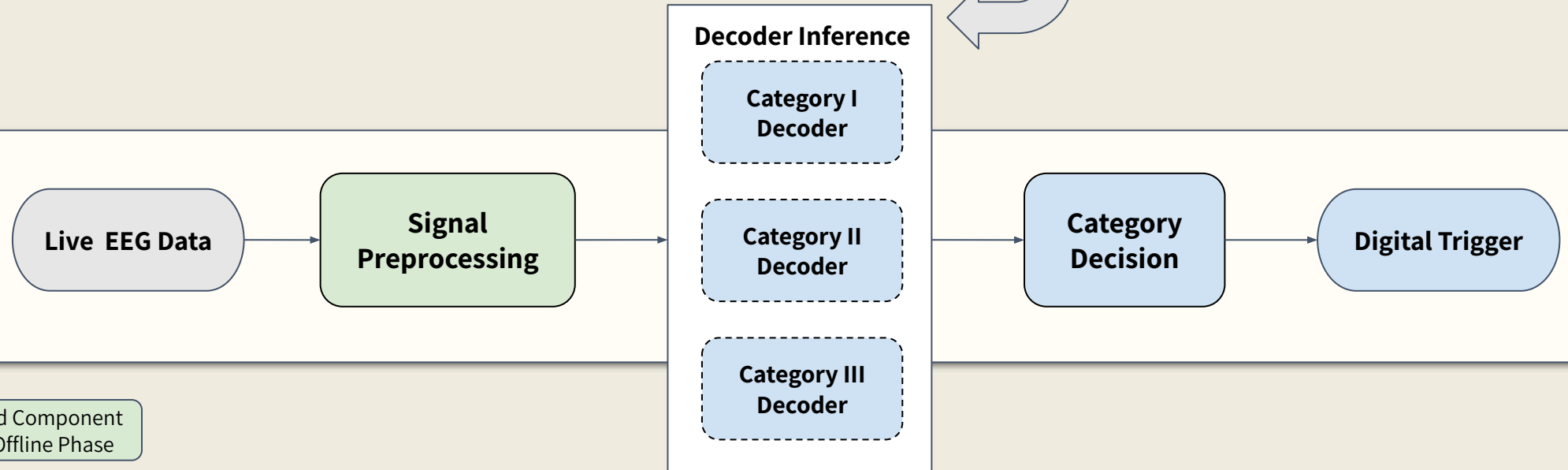
Experiment Phase 1

Collecting live data and training specific decoders per category for current subject



Experiment Phase 2

Using trained decoders to Identify memory reactivation on real-time EEG data



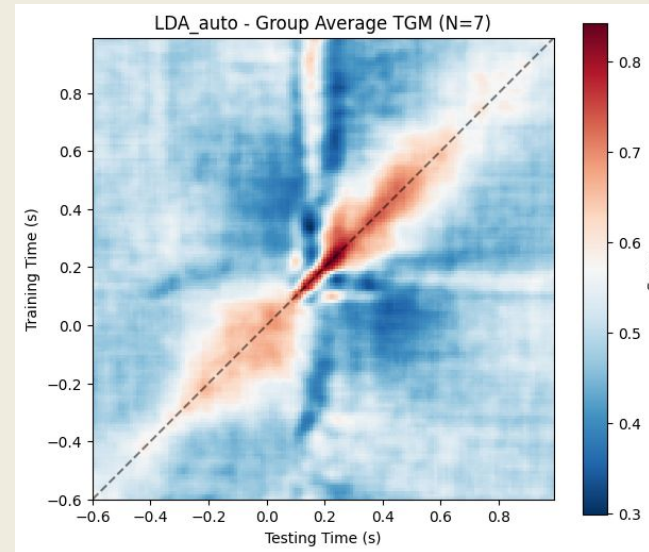
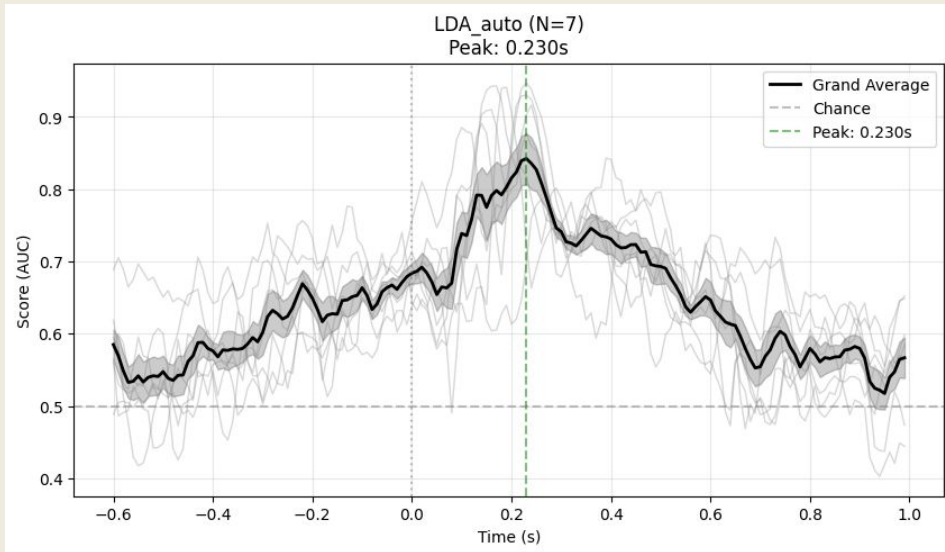
Project Components

Adapted Component
From Offline Phase

All components to be implemented during semester B

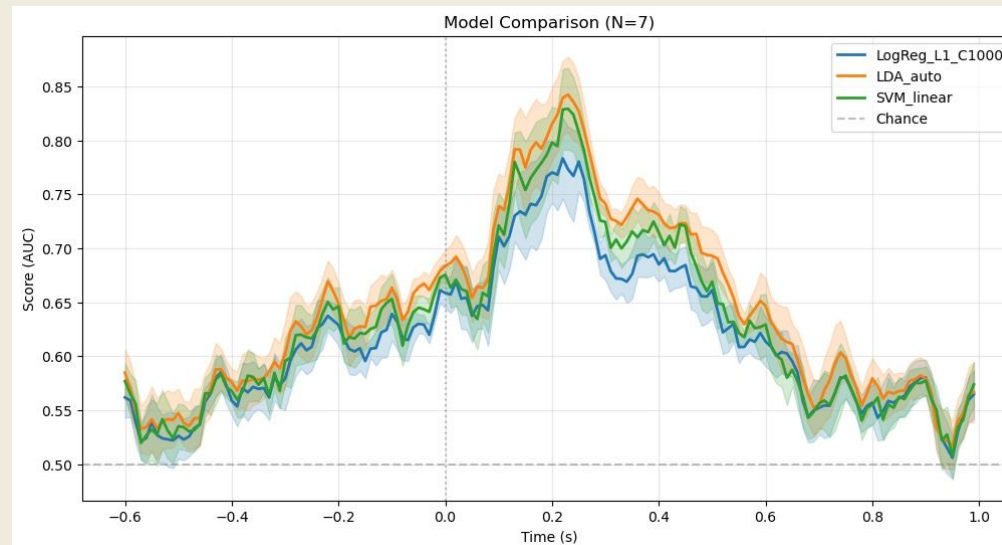
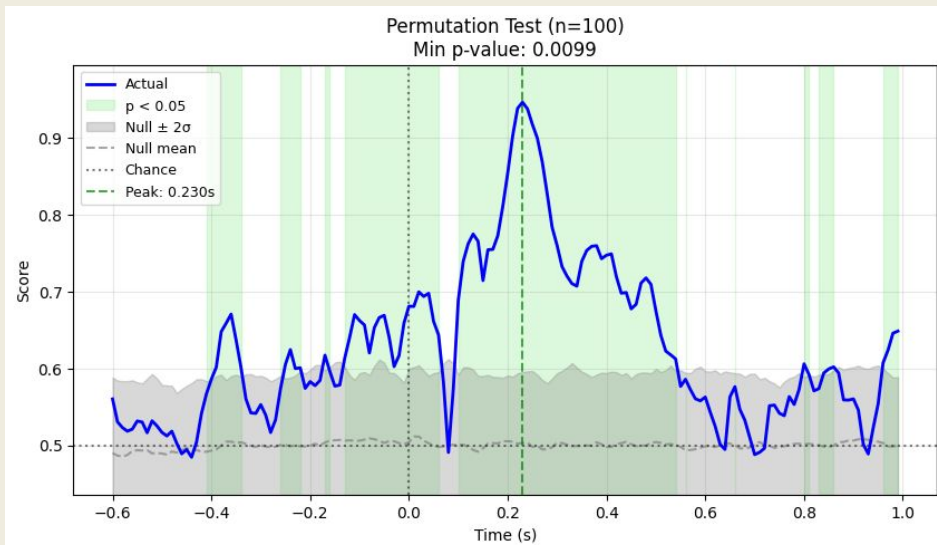


Achieved Results – Classification

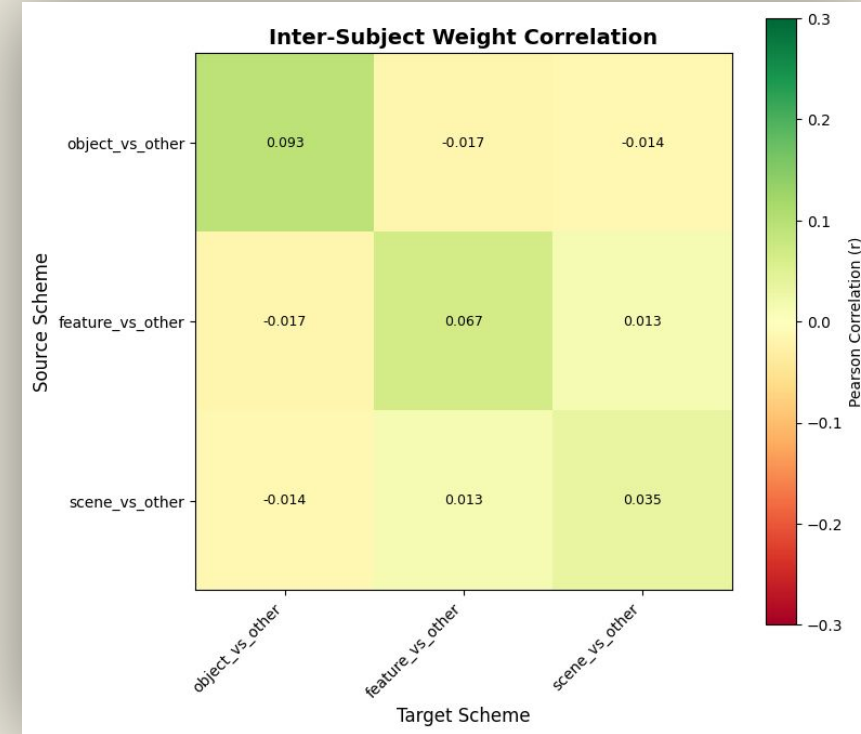
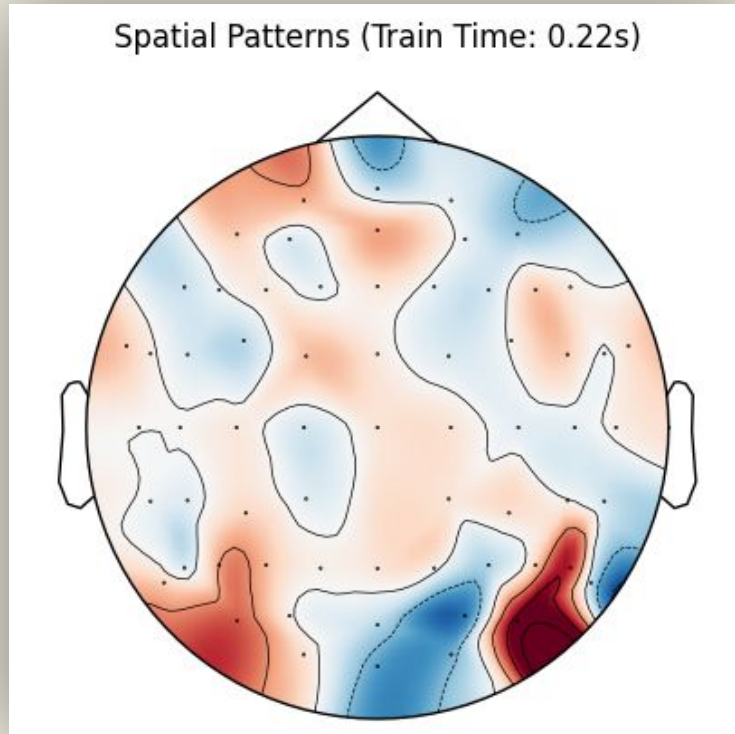


Offline Decoder Performance Evaluation:

- Achieved Performance of 80% object classification at peak decoding time (avg across subjects)
- Achieved p-value < 0.05 at 230ms (peak)

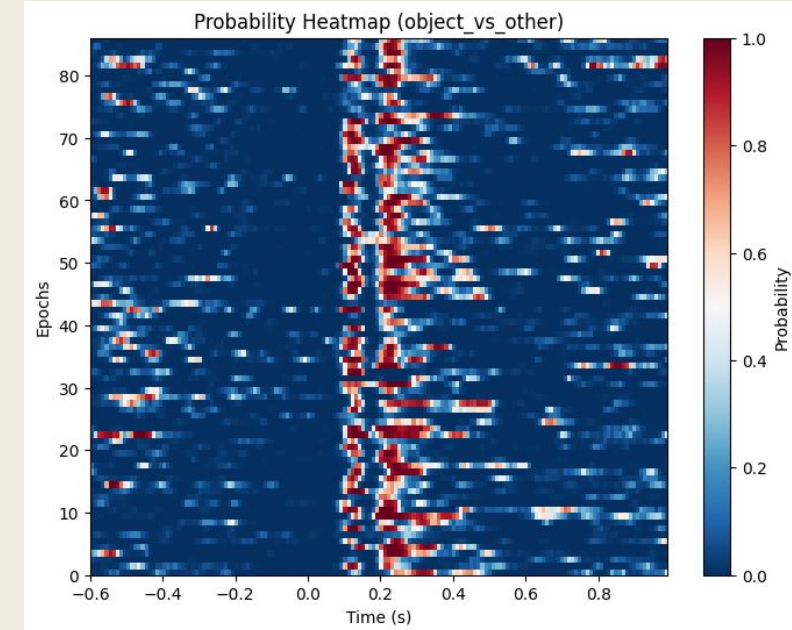
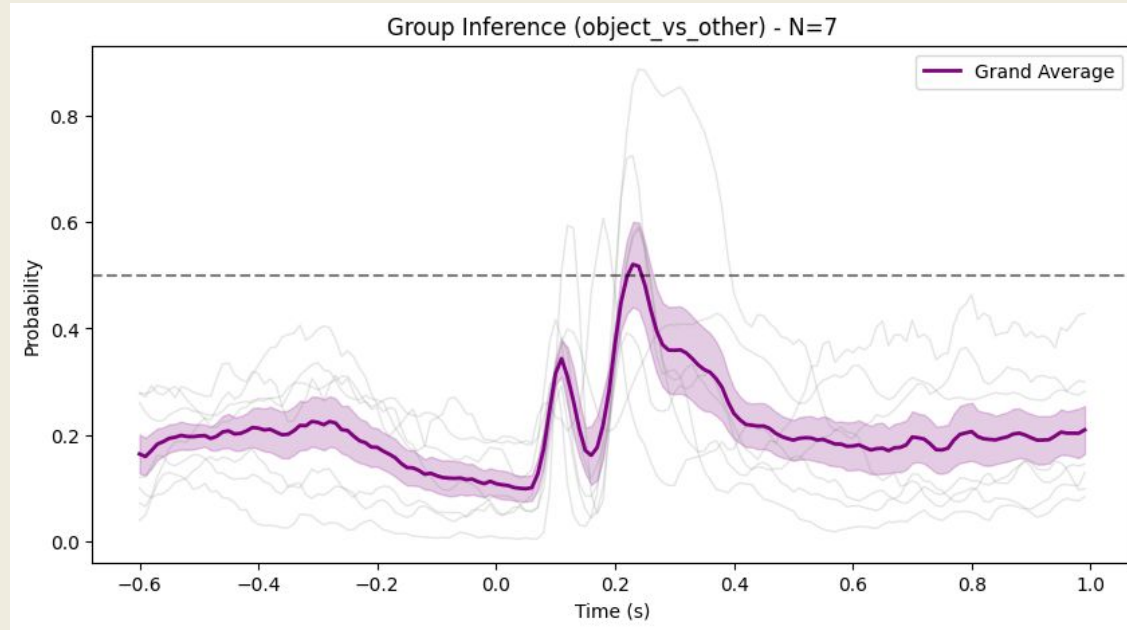


Achieved Results – Explainability



- The spatial pattern suggests that the occipital channels impacted the decoding the most, which makes sense for visual memory reactivation.
- We can see stronger correlation when comparing the same category across subjects (diagonal) - confirms that the EEG patterns are unique to the stimulus content.

Achieved Results – Inference



Inference Results:

- Achieved above chance level on retrieval epochs for object only.
- The plan to collect new data is better suited for our specific goal and therefore will yield better results.

Updated Schedule

So Far

Milestone	Planned	Actual Date	Comments
Initial Literature Review	16.11.25	On time	
Workplan submission	16.11.25	On time	
Data Familiarity	30.11.25	On time	
Implementing Preprocessing Pipeline (D1)	14.12.25	Not relevant	Used Existing
Offline Model Training	25.01.26	On time	
Binary Model Evaluation & Explainability (D2, D3)	08.02.26	On time	
Creating a package for offline evaluation and inference	[NEW]	08.02.26	
Go/No-Go Decision for New Data Acquisition	08.02.26	On time	New Experiment planned for improved data
Progress Presentation Submission	22.02.26	On time	Determined by presentation date

Next Steps

Milestone	Planned
Requirements and interface document for the online system	15.03.26 [NEW]
Real-Time Pipeline Development	20.04.26
Online Simulation & Validation (D5)	30.04.26
(Optional) Live Pilot Testing	18.05.26
Poster Submission and finishing the work	24.05.26
Online System UI	15.06.26
Final Documentation & Report Writing	28.06.26
Final deliverables submission	01.07.26



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Questions?